Fall 2025 ECEN 629 Homework Set 2

Due Sep. 30th 12:45pm online, or 2:00pm in class

11.1%*9

Prob. 1. Suppose $f : \mathbf{R} \to \mathbf{R}$ is concave, and $a, b \in \mathbf{dom} f$ with $a < b$.

(a) Prove the following inequality:

$$\frac{f(x) - f(a)}{x - a} \geq \frac{f(b) - f(a)}{b - a} \geq \frac{f(b) - f(x)}{b - x}.$$

for any $x \in (a, b)$.

(b) For $f = -x^2 + 2x + 5$, $a = 0$, and $b = 3$, draw a sketch to illustrate the inequalities above.

Solution:

(a) The first inequality is equivalent to

$$f(x) \geq \frac{x - a}{b - a} f(b) + \frac{b - x}{b - a} f(a).$$

However, notice that $\frac{x-a}{b-a} b + \frac{b-x}{b-a} a = x$, thus this inequality is simply Jensen's inequality. The second inequality can be rewritten similarly in another form.

If the function is twice differentiable, then letting $x \to a$ or $x \to b$, we have $f'(a) \geq (f(b) - f(a))/(b - a) \geq f'(b)$. We know the $f'(x)$ is monotonically non-increasing ($f''(x) \leq 0$), i.e., $f'(a) \geq f'(b)$, since $f(x)$ is a concave function; moreover, rewriting the second equality gives $f(b) + f'(b)(a - b) \geq f(a)$, which is precisely the first order condition for concave functions at point b (a negation of a convex function).

(b) See next page. The plot can be interpreted as the slope of the secant line through $(a, f(a))$ and $(x, f(x))$ is larger than the slope of the secant line through $(a, f(a))$ and $(b, f(b))$, which in turn is larger than the slope of the secant line through $(x, f(x))$ and $(b, f(b))$.

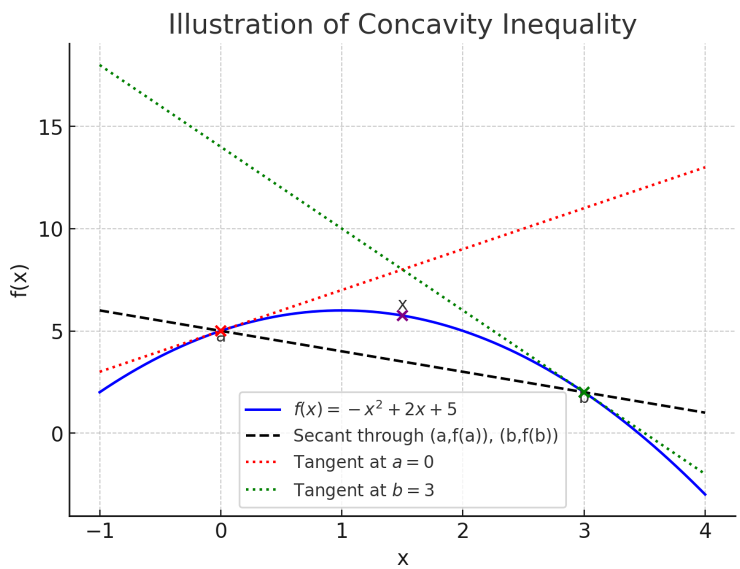
This reflects the fact that for a concave function, the slope of the tangent (derivative) is non-increasing:

$$f'(a) \ \geq \ \frac{f(b) - f(a)}{b - a} \ \geq \ f'(b).$$

Prob. 2. Show that the maximum of a convex function f over the polyhedron $\mathcal{P} = \mathbf{conv}\{v_1, v_2, \ldots, v_k\}$ is achieved at one of the vertices, that is

$$\sup_{x \in \mathcal{P}} f(x) = \max_{i=1,2,\ldots,k} f(v_i).$$

In particularly, let $f(x) = x_1^2 + 2x_1x_2 + 3x_2^2$, and let $v_1 = (1, 1)^T$, $v_2 = (1, -1)^T$, $v_3 = (-1, 1)^T$, and $v_4 = (-1, -1)^T$. Compute its maximum over the polyhedron.



hb!

Solution: Suppose this is not true, then there is a $z \in \mathcal{P}$ such that $f(z) > f(v_i)$, $i = 1, 2, ..., k$. By definition of the convex hull we must have $z = \lambda^T v$, where $\lambda \geq 0$, and $1^T \lambda = 1$. It follows that

$$f(z) = f(\lambda^T v) \leq \lambda^T f(v) < f(z),$$

which is a contradiction.

We can compute the values at these four points, and find the maximum $f(v_1) = f(v_4) = 6$.

Prob. 3. Suppose

$$f(x) = h\big(g_1(x), g_2(x)\big),$$

where $g_1(x) = \|x\|_2^2$, $g_2(x) = \|x\|_1$, and $h(u, v) = \log(e^u + e^v)$.

(a) Show $f(x)$ is convex by applying the correct composition rule.

(b) Give another h that keeps f convex but is non-logarithmic, and explain why.

Solution: We prove a more general result: Let

$$f(x) = h(g_1(x), g_2(x), \ldots, g_k(x))$$

where $h : \mathbf{R}^k \to \mathbf{R}$ is convex, and $g_i : \mathbf{R}^n \to \mathbf{R}$. Suppose that for each i, one of the following holds:

- h is nondecreasing in the i-th argument, and g_i is convex;
- h is nonincreasing in the i-th argument, and g_i is convave;
- g_i is affine.

Then the function f is convex. This can be proved as follows: We can reorder the g_i's such that the first p of them is affine, the next $q - p$ are convex, and the last $k - q$ are concave. Let $z = \theta x + (1 - \theta)y$, then because the monotonicity of h in its separate arguments

$$f(z) = h(g_1(z), g_2(z), \ldots, g_k(z)) \leq h(\theta g_1(x) + (1 - \theta)g_1(y), ..., \theta g_k(x) + (1 - \theta)g_k(y)).$$

Because h is convex, we continue to write

$$f(z) \leq \theta h(g_1(x), ..., g_k(x)) + (1-\theta) h(g_1(y), ..., g_k(y)) = \theta f(x) + (1-\theta) f(y).$$

Prob. 4. Let $f_1, f_2, \ldots, f_m$ be convex function on $\mathbf{R}^n$. Define

$$f(x) = \inf\{f_1(x_1) + f_2(x_2) + \ldots + f_m(x_m) | x_i \in \mathbf{R}^n, x_1 + x_2 + \ldots + x_m = x\}.$$

Show $f(x)$ is convex on $\mathbf{R}^n$.

Solution: We prove this by definition. For simplicity, let us assume the inf can be attended, i.e.,

$$\begin{aligned}
&\theta f(y) + (1-\theta) f(y') \\
&= \theta \cdot \inf[f_1(x_1) + f_2(x_2) + \ldots + f_m(x_m)] + (1-\theta) \cdot \inf[f_1(x_1') + f_2(x_2') + \ldots + f_m(x_m')] \\
&= \theta[f_1(x_1^*) + f_2(x_2^*) + \ldots + f_m(x_m^*)] + (1-\theta)[f_1(x_1'^*) + f_2(x_2'^*) + \ldots + f_m(x_m'^*)]
\end{aligned}$$

where $x_1^* + x_2^* + \ldots + x_m^* = y$, and $x_1'^* + x_2'^* + \ldots + x_m'^* = y'$. Then it follows that

$$\begin{aligned}
&\theta[f_1(x_1^*) + f_2(x_2^*) + \ldots + f_m(x_m^*)] + (1-\theta)[f_1(x_1'^*) + f_2(x_2'^*) + \ldots + f_m(x_m'^*)] \\
&\geq f_1(\theta x_1^* + (1-\theta) x_1'^*) + f_2(\theta x_2^* + (1-\theta) x_2'^*) + \ldots + f_m(\theta x_m^* + (1-\theta) x_m'^*) \\
&\geq \inf[f_1(x_1'') + f_2(x_2'') + \ldots + f_m(x_m'')],
\end{aligned}$$

where the infimum is taken over $(x_1'', x_2'', \ldots, x_m'')$ such that

$$x_1'' + x_2'' + \ldots + x_m'' = \theta y + (1-\theta) y',$$

i.e., we have $\theta f(y) + (1-\theta) f(y') \geq f(\theta y + (1-\theta) y')$. When the infimum is not attended, we will need to take a sequence of points that approaches it, however, the argument remains the same.

Prob. 5. The r-separation function $\Delta_r(x)$ of the vector $x \in \mathbf{R}^n$, where $r \leq n$, is defined as the difference between the sum of the r largest components of the vector x and the sum of the r smallest components.

(a) Is the function $\Delta_r(x)$ convex? Prove your answer.

(b) Let $r = 1$ and $n = 2$. Sketch the function $\Delta_1(x)$ in the positive quadrant.

Solution:

(a) Define the following functions

$$f_{R_1,R_2}(x) = \sum_{i \in R_1} x_i - \sum_{j \in R_2} x_j,$$

where R_1, R_2 are subsets of $\{1, 2, \ldots, n\}$ whose cardinalities are r. Then

$$\Delta_r(x) = \max_{R_1, R_2} f_{R_1,R_2}(x).$$

Since $f_{R_1,R_2}(x)$ is clearly convex, this point-wise maximum function is convex.

(b) See attached sketch.

Prob. 6. Suppose $f : \mathbf{R}^m \to \mathbf{R}$ is convex, and $A \in \mathbf{R}^{m\times n}$, $b \in \mathbf{R}^m$, $c \in \mathbf{R}^n$, and $d \in \mathbf{R}$. Show that the function that $g : \mathbf{R}^n \to \mathbf{R}$ is convex, where

$$g(x) = (c^T x + d) f\left(\frac{Ax + b}{c^T x + d}\right), \quad \mathbf{dom}\, g = \{x | c^T x + d > 0\}.$$

Provide a clear argument which rules you are applying, and the exact manner of function composition.

Solution: Recall the definition of the perspective function of function $f(\cdot)$ is $\tilde{f}(x,t) = tf(x/t)$, and thus $g(x)$ is the composition of $\tilde{f}(x,t)$ with another affine transform $x \to (Ax + b, c^T x + d)$. By the composition rule with affine functions, $f(x)$ is also convex.

Prob. 7. A vector-valued function $f : \mathbb{R}^n \to \mathbb{R}^n$ is called monotone if $\langle f(x) - f(y), x - y\rangle \geq 0$. Show that the gradient of a convex function is monotone.

Solution: From the first order condition we have

$$f(y) \geq f(x) + \nabla f(x)^T (y - x),$$

and at the same time

$$f(x) \geq f(y) + \nabla f(y)^T (x - y).$$

Add the two inequalities together, and we arrive

$$[\nabla f(x)^T - \nabla f(y)^T](x - y) = \langle \nabla f(x) - \nabla f(y), x - y\rangle \geq 0,$$

which is the desired inequality.

Prob. 8. The proximal operator of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is defined by

$$\mathrm{prox}_{\lambda f}(v) = \arg\min_x \left(f(x) + \frac{1}{2\lambda}\|x - v\|_2^2 \right).$$

(a) Is the optimization problem to find the proximal operator convex?

(b) Derive the proximal operator for $f(x) = \|x\|_1$.

(c) Solve explicitly for $v = (3, -2, 0.5)^T$ and $\lambda = 1$ when $f(x) = \|x\|_1$.

Solution:

(a) $x \mapsto \frac{1}{2\lambda}\|x - v\|_2^2$ is convex for any $\lambda > 0$, and if $f(x)$ is convex, then the sum of convex functions is convex. If $f(x)$ is not convex, it may still be possible that the problem is convex, if we choose λ sufficiently small.

(b) The proximal operator of the ℓ_1 norm is

$$\arg\min_x \left(\|x\|_1 + \tfrac{1}{2\lambda}\|x - v\|_2^2 \right).$$

This problem decouples coordinate-wise. For each coordinate i we solve

$$\min_{x_i \in \mathbb{R}} \ |x_i| + \tfrac{1}{2\lambda}(x_i - v_i)^2.$$

- If $x_i > 0$: The objective is $x_i + \frac{1}{2\lambda}(x_i - v_i)^2$. Taking the derivative, gives $x_i = v_i - \lambda$. This is valid if $v_i > \lambda$. Therefore, if $v_i > \lambda$, then the optimal point is $x_i^* = v_i - \lambda > 0$.
- If $x_i < 0$: The objective is $-x_i + \frac{1}{2\lambda}(x_i - v_i)^2$. Derivative condition gives $x_i = v_i + \lambda$. This is valid if $v_i < -\lambda$. Therefore, if $v_i < -\lambda$, then the optimal point $x_i^* = v_i + \lambda < 0$.
- If $x_i = 0$: We consider the remaining case when $v_i \in [-\lambda, \lambda]$. The question is whether $x_i = 0$ is indeed a minimizer in this case. Mathematically, this requires an argument involving the subgradient, but here we can easily verify that when $v_i \in [-\lambda, \lambda]$, the minimizer can only happen in $[0, v_i]$ if $v_i \geq 0$ (and in $[v_i, 0]$ if $v_i < 0$). In the first case, it can be easily verified that the function $x_i + \frac{1}{2\lambda}(x_i - v_i)^2$ is monotonically increasing, since the derivative is between 0 and 1. Therefore, indeed $x_i = 0$ is a minimizer. The other case is similar.

Combining these cases,

$$x_i^\star = \operatorname{sign}(v_i)\ \max\{|v_i| - \lambda, 0\}.$$

Therefore, the proximal operator is the *soft-thresholding operator*:

$$\operatorname{prox}_{\lambda\|\cdot\|_1}(v) = \big(\operatorname{sign}(v_i)\ \max\{|v_i| - \lambda, 0\}\big)_{i=1}^n$$

(c) With $v = (3, -2, 0.5)^\top$ and $\lambda = 1$,

$$\begin{aligned}
&\operatorname{prox}_{\|\cdot\|_1}(v) \\
&= \big(\operatorname{sign}(3)\max\{3-1,0\},\ \operatorname{sign}(-2)\max\{2-1,0\},\ \operatorname{sign}(0.5)\max\{0.5-1,0\}\big) \\
&= (2,\ -1,\ 0)^\top.
\end{aligned}$$

Prob. 9. Given a norm $\|\cdot\|$ on $\mathbb{R}^n$, the dual norm is defined as

$$\|y\|_* = \sup_{\|x\| \leq 1} y^T x.$$

(a) Show that $\|\cdot\|_*$ is indeed a norm.

(b) Compute the dual norm of $\|x\|_1$ and $\|x\|_\infty$.

(c) For $x = (1, 2)^T$, compute $\|x\|_2^*$.

Solution:

(a) Let $\|y\|_* = \sup_{\|x\| \leq 1} y^\top x$. We verify the norm axioms:

- Nonnegativity and definiteness: Clearly $\|y\|_* \geq 0$. If $\|y\|_* = 0$, then $y^\top x = 0$ for all $\|x\| \leq 1$. Taking $x = y/\|y\|$ if $y \neq 0$ gives a contradiction, hence $y = 0$.
- Homogeneity: $\|\alpha y\|_* = \sup_{\|x\| \leq 1} (\alpha y)^\top x = |\alpha| \sup_{\|x\| \leq 1} y^\top x = |\alpha| \|y\|_*$.
- Triangle inequality: $\|y + z\|_* = \sup_{\|x\| \leq 1} (y + z)^\top x \leq \sup_{\|x\| \leq 1} y^\top x + \sup_{\|x\| \leq 1} z^\top x = \|y\|_* + \|z\|_*$.

Thus $\|\cdot\|_*$ is a norm.

(b)
- For $\|x\|_1$, the dual is

$$\|y\|_* = \sup_{\|x\|_1 \leq 1} y^\top x \leq \|y\|_\infty,$$

with equality attained at $x = e_j \operatorname{sign}(y_j)$ where $|y_j| = \|y\|_\infty$. Hence $(\|\cdot\|_1)^* = \|\cdot\|_\infty$.

- For $\|x\|_\infty$, the dual is

$$\|y\|_* = \sup_{\|x\|_\infty \le 1} y^\top x \le \sum_i |y_i| = \|y\|_1,$$

  with equality attained at $x_i = \mathrm{sign}(y_i)$. Hence $(\|\cdot\|_\infty)^* = \|\cdot\|_1$.

(c) For $x = (1,2)^\top$ under ℓ_2, since the dual of ℓ_2 is itself:

$$\|x\|_2^* = \|x\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}.$$